INCH-POUND
MIL-PRF-1/1386F
2 December 2008
SUPERSEDING
MIL-E-1/1386E
20 July 1976

PERFORMANCE SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPE 4QP2

Inactive for new design
after 5 December 1997

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the electron tube described herein shall consist of this document and the latest issue of MIL-PRF-1.

DESCRIPTION: Flat face, rectangular, electrostatic deflection and focus, post accelerator.

DIMENSIONS AND PIN CONNECTORS:- See figure 1.

ABSOLUTE RATING:

Parameter:	Ef	Ec1	Eb1	Eb2	Eb3	Ed	Rg
Unit:	V	Vdc	Vdc	Vdc	Vdc	V	Meg Ω
Maximum	6.9	0	160	650	3,400	200	1.5
Minimum:	5.67	-200	---				
TEST CONDITIONS: (Cathode reference) (Notes 1 and 2)	6.3	Adjust	Focus	575	3,000	----	-----
TEST CONDITIONS: (Oscilloscope use) (Ground reference) (Notes 1 and 2)	6.3	Adjust	Focus	-----	-----	----	----

ABSOLUTE RATINGS:

Parameter:	Zd	Ehk	Eb3/Eb2	Ek	Egeom	Esd	East	Escn
Unit:	Meg Ω (Note 2)	Vdc	Ration	Vdc	Vdc	Vdc	Vdc	Vdc
Maximum	----	+180	5.2	-----	-----	----	----	-----
Minimum:	1.0	-180	----	----	-----	-----	-----	-----
TEST CONDITIONS: Cathode reference) (Notes 1 and 2)	---------	-----	----	------	-----	----	-----	-----
TEST CONDITIONS: (Oscilloscope use) (Ground reference) (Notes 1 and 2)	----------	-----	----	-575	Adjust	Adjust	Adjust	2,425

GENERAL:
Qualification - Required

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TABLE I. Testing and inspection.

Inspection	MIL-STD-1311 Method	Notes	Conditions	Symbol	Limits		Unit
					Minimum	Maximum	
Conformance inspection, Part 1							
Temperature cycling	1027	-		-----	5	-----	Cycles
Pressure (implosion)	1141	-		-----	-----	-----	-----
Barometric pressure, reduced	1002	-		-----	-----	-----	-----
Low-temperature operation	1026	-		-----	-55	-----	°C
Vibration	5111	-		-----	-----	1	mm
Cathode illumination	5216	-		-----	-----	-----	----
Direct-interelectrode capacitance	1331	-	Cathode to all	Ck	-----	5.3	pF
			Grid No. 1 to all	Cg1	-----	6.8	pF
			D1 to D2	C1D2	-----	2.8	pF
			D3 to D4	C3D4	-----	1.0	pF
			D1 to all	CD1	-----	6.3	pF
			D2 to all	CD2	-----	6.7	pF
			D3 to all	CD3	-----	3.8	pF
			D4 to all	CD4	-----	3.8	pF
Neck and bulb alinement (electrostatic types)	5101	-	----	-----	------	------	------
Face tilt	5101	-				0.080	Inch
Deflection-factor uniformity	5248	8	1D2	DFU	-----	3	%
			3D4	DFU	-----	3	%
Magnetization	5256	-		----	-----	-----	----
Shock	-----	0		------	---	----	------
Conformance inspection, part 1		12					
Electrode current (anode No. 3)	5201	3		---	-----	15	µAdc
Voltage breakdown	5201	-		---	-----	---	-----
Voltage breakdown (electrostatic types)	5201	-		---	-----	-----	-----
Gas “cross”	5206	13	Ib3-15µA	---	-----	-----	-----
Angle between traces	5101	-			89	91	Degrees
Screen and faceplate blemishes	5106	-			-----	-----	-----
Modulation	5223	-	Ib3 = 10 µA	ΔEc1	-----	30	Vdc

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TABLE I. Testing and inspection. Continued

Requirement or Test	MIL-STD-1311 Method	Notes	Conditions	Symbol	Limits		Unit
					Minimum	Maximum	
Conformance inspection, Part 1 Continued							
Line with “A” (electrostatic deflection)	5226	4	Ib3 = 10 µA	-----	-----	0.75	mm
Line width “B” (electrostatic deflection)	5226	---	Ib3 = 10 µA	-----	----	0.80	mm
Spot postion (electrostatic deflection)	5231	---	ID2 3D4	----- -----	---- ----	6 8	mm mm
Spot displacement (leakage)	5241	---		-----	----	10	mm
Grid cutoff voltage	5241	---		Ec1	-40	-20	Vdc
Focusing voltage at cutoff	5246	---		Eb1	----	180	Vdc
Deflection factor	5248	---		DF DF	22.5 13.5	27.5 16.5	Vdc/in. Vdc/in.
Grid No. 1 leakage current	5251	---		----		3	µAdc
Anode No 1 leakage current	5251	----		----		5	µAdc
Anode No. 2 leakage current	5251	---		----		5	µAdc
Geometry adjust	-----	6, 10		----	520	595	Vdc
Deflection plate shield	-----	7, 10	3D4	----	520	595	Vdc
Astigmatism adjust	-----	10		----	520	595	Vdc
Useful scan	-----	--	1D2 3D4	---- ----	---- ----	2.50 2.25	Inch Inch
Pattern distortion	5			----	----	----	-----
Conformance inspection, Part 2							
Heater current	1301	--	Ec1 = 2 Vdc	If	540	660	mA
Electrode current (anode No. 1)	5201	---	Ib3 = 15µA	Ik	-15	+10	µAdc
Electrode current (cathode)	5201	---					
Base alinement (electrostatic types)	5101	---	+ID2,Pin No. 4	------	----	0.500	mAdc
Side terminal and base alinement (electrostatic types)	5101	---	+ID2	-----	-----	----	----

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TABLE I. Testing and inspection. Continued

Requirement or Test	MIL-STD-1132 Method	Notes	Conditions	Symbol	Limits		Unit
					Minimum	Maximum	
Conformance inspection, Part 2 Continued							
Side terminal and base alinement	5101	---	Pin No. 4		---		
High-temperature operation	1026	11			-----	+70	°C
Neck and base alignment (electrostatic types)	5101	----		-----	------	-----	-----
Stray light emission (conventional types)	5216	----	Eb2 = 600 Vdc; Eb3 = 3,400 Vdc	--- ----	---- ----	---- ----	----
Screen	5221	----		----	360	----	cb
Focusing voltage, zero-bins	5246	----		Eb1	40	----	Vdc
Heater-cathode leackage current	5251	--		Ihk	----	10	µAdc
Secureness of base, cap, or insert	1101	----		----	----	----	----
Permanence of marking	1105	----		----	----	----	----
Conformance inspection, Part 3							
Life test			Group C; Ib3 = 10 µAdc; Eb2 = 575 Vdc; Eb3 = 3,000 Vdc	T	500	----	hrs
Life-test end points:							
Line width “A”	5226	----	Ib3 = 10µA	----	----	0.75	mm
Line width “B’	5226	----	Ib3 = 10µA	----	----	0.80	mm
Modulation	5223	----	Ib3 = 10µA	ΔEc1	----	30	Vdc

NOTES:

1. All readings of beam current (Ib3) shall be in addition to the reading obtained for method 5201, anode No. 3 current. For oscilloscope use, the mean potentials of 1D2 and 3D4 with respect to ground shall be zero.
2. It is recommended that the deflecting electrode circuit resistance be approximately equal. Higher resistance values up to 5.0 megohms may be used for low-beam current operation.
3. With the beam cutoff, the post-accelecrator anode (A3) shall draw 15 µAdc maximum.
4. Spot adjusted for least astigmatism.
5. With a 2.5-by 2-inch rectangular raster centered on the face of the tube, the raster edge shall not deviate form straight parallel lines by more than 0.040 inch on the left and right edges nor by more than 0.0515 inch at the top and bottom.

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6. The post-accelerator spiral band lower end and the geometry adjust are connected internally. By adjusting the potential on this electrode combination, pin cushion and barrel distortions are minimized.

7. Normally operated at accelerator potential, linearity improvements can be obtained in the vertical trace by proper adjustment of deflection plate shield voltage.

8. For the 102 direction: Divide the axis into five successive 1/2 –inch intervals symmetrically located with respect to the geometric center of the tube face. The percentage difference (DFU)between deflection factors of the center interval and any other interval shall not exceed the limit specified.

9. A hammer-drop shock testing machine shall be used, consistent with MIL-PRF-28800. The tube may be clamped as in the vibration test or mounted directly on the shock machine. With test potentials applied, display a circle or raster. Using a hammer-drop height of 6 inches, apply three blows each in directions 1D2, 3D4, and base-to face. Momentary failure of operation during shock is permissible provided normal operation is deflection factor, spot position, and grid cutoff voltage. In addition, there shall be no evidence of damage to the internal structure.

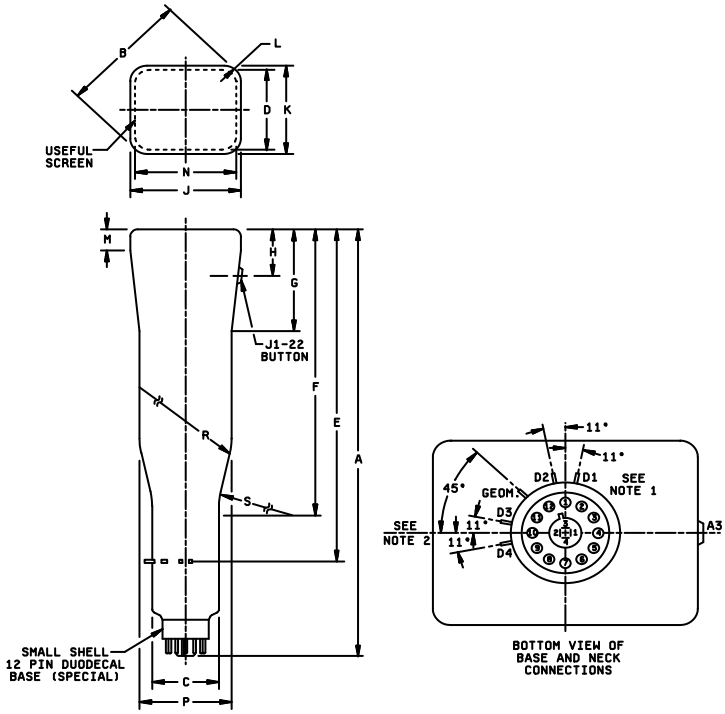
10. Range for oscilloscope use: -55 to +20 Vdc.

11. Following temperature stabilization, the tube shall meet requirements under pattern distortion, line-width, and anode No. 3 current; all voltages and settings being those obtained at room temperature with 25 K resistors in series with geometry, shield and astigmatism adjust electrodes.

12. An accept on zero defect sampling plan (see MIL-PRF-1, table III) shall be used for conformance inspection, part 1.

13. This test to be performed at the conclusion of the holding period.

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PIN NO.	ELEMENT
1	HEATER
2	CATHODE
3	GRID
4	N.C.
5	AI
6	N.C.
7	AZ
8	N.C.
9	V SHELD
10	N.C.
11	N.C.
12	HEATER

FIGURE 1. Outline drawing of electron tube types 4QP2

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Ltr	Dimensions are in inches with metric equivalents (mm) in parentheses.	
	Mimimum	Maximum
Conformance inspection, part 2		
A	13.312 (338.12)	13.688 (347.68)
B	- - - - - - - -	3.765 (95.63)
C	1.969 (50.01)	2.031 (51.59)
D	2.250 (57.15)	- - - - - - -
E	10.000 (254.00)	10.250 (260.35)
H	1.250 (31.75)	1.500 (38.10)
J	3.219 (81.76)	3.281 (83.34)
K	2.719 (69.06)	2.781 (70.64)
L	- - - - - - -	.500 (12.70)
P	2.719 (69.06)	2.781 (70.64)
Reference dimensions (see note 3)		
F	8.500 (215.90)	
G	3.125 (79.38)	
M	.625 (15.88)	
R	5.500 (139.70) Rad.	
S	3.000 (76.20) Rad.	

NOTES:
1. +1D2 towards pin No. 4.
2. +3D4 towards pin No. 1.
3. These dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing of electron tube types 4QP2 - Continued

Referenced documents. In addition to MIL-PRF-1, this document references the following:
MIL-PRF-28800
MIL-STD-1311

Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Navy – EC
DLA – CC

Preparing activity:
DLA- CC

Project number (5960-2008-071)

Review activities:
Navy – AS, CG, MC, OS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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